Activation of the JNK signaling pathway: breaking the brake on apoptosis

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Summary
The JNK signaling pathway is involved in regulation of many cellular events, including growth control, transformation and programmed cell death (apoptosis). The role of JNK activation in apoptosis is highly controversial, being suggested to have a pro-apoptotic, anti-apoptotic or no role in this process. It appears that the JNK pathway functions in a cell-type and stimulus-dependent manner and its different components can sometimes play opposing roles in apoptosis. Recent studies reveal that the effect of JNK activation on apoptosis depends on the activity of other signaling pathways like the NF-κB pathway. Here we propose a model that can explain how activation of the JNK pathway "breaks the brake" on apoptosis, thereby regulating, but not initiating the apoptotic process. *BioEssays* 25:17–24, 2003.
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Introduction

Programmed cell death (apoptosis) is essential to many biological processes in multicellular organisms, including embryonic development, immune responses, tissue homeostasis and normal cell turnover.[1–4] While the initiation and execution of apoptosis depend on activation of the receptor- and/or mitochondrial-dependent death pathways,[5–8] the apoptotic process is affected by many other signaling pathways including the JNK pathway.[9–12]

c-Jun N-terminal protein kinase (JNK) was originally identified as two protein kinases of $p46^{JNK1}$ and $p54^{JNK2}$,[13] which specifically phosphorylate the transcription factor c-Jun on its N-terminal transactivation domain at Ser63 and Ser73.[13–15] This phosphorylation can be induced by a variety of extracellular stimuli,[13–17] resulting in the augmentation of c-Jun transcriptional activity and its ability to cooperate with *Ha*-Ras in transformation.[16,17] Thus, JNK is a key regulator of c-Jun activity.[18] Molecular cloning of JNK[19] (also known as stress-activated protein kinase (SAPK)[20] revealed that it is a member of the mitogen-activated protein kinase (MAPK) superfamily, having three isoforms (JNK1, JNK2 and JNK3) with slicing variant.[10] JNK has been reported to activate transcription factors other than c-Jun, such as ATF2, Elk-1, p53 and c-Myc[10,18] and non-transcription factors, such as members of the Bcl-2 family (Bcl-2 and Bcl-x_L).[21,22]

JNK is activated by sequential protein phosphorylation through a MAP kinase module, i.e., MAP3K → MAP2K → MAPK[18] (Fig. 1). Two MAP2Ks (JNKK1/MKK4/SEK1 and JNKK2/MKK7) for JNK have been identified.[23–27] These dual-specificity protein kinases phosphorylate JNK on Thr183 and Tyr185, resulting in JNK activation. Interestingly, these two JNKKs have different preferences for the phosphate-acceptor sites, with JNKK1 preferring Tyr185[23,28] and JNKK2 Thr183.[28] Although both JNKK1 and JNKK2 may be required for full activation of JNK, the differential phosphorylation of JNK by JNKKs may provide a molecular basis for activation of JNK by various stimuli. Several MAP3Ks, including members of the MEKK family,[30] ASK1,[30] MLK,[31] TAK1[32] and TPL-2,[33] have been reported to act as MAP3Ks for JNK. While genetic disruption of MEKK1 in mice abrogates JNK activation by certain stimuli,[34,35] the physiological relevance of other MAP3Ks in JNK activation is less clear. Activation of JNK is also regulated by scaffold proteins such as JIP, β-arrestin and JSAP1.[36–38] Termination of JNK activation can be achieved by protein phosphatases such as mitogen-activated protein phosphatases.[18] More recently, studies have revealed that activation of JNK can be negatively regulated by NF-κB-mediated inhibition, as will be discussed below.[39–42]

The function and regulation of the JNK pathway has been the subject of several recent reviews.[10–12] This review will focus on emerging concepts about the role of the JNK pathway in apoptosis. We also propose a novel model that can explain promotion of TNF-α-induced apoptosis by prolonged JNK activation in the absence of NF-κB.

The JNK pathway and apoptosis: a complex story

Numerous studies have implicated both pro-apoptotic and anti-apoptotic functions for the JNK pathway.[43–45] Evidence

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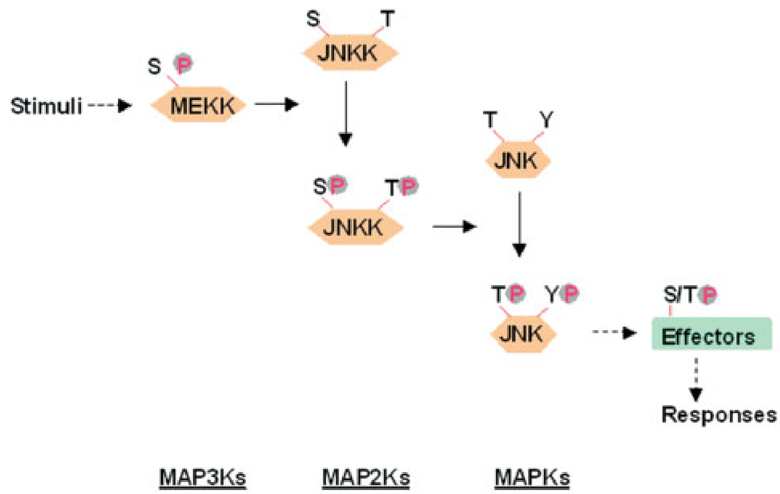

Figure 1. Schematic illustration of activation of the JNK MAP kinase module. JNK is activated by sequential protein phosphorylation through the MAP kinase module, in which MAP3Ks like MEKK phosphorylates and activates dual specificity MAP2Ks such as JNKK. JNKK in turn phosphorylates JNK on both threonine and tyrosine, thereby stimulating JNK's activity. Activated JNK phosphorylates downstream effectors, resulting in changes in patterns of gene expression and other cellular responses.

also exists that JNK activation plays no role in the apoptotic process.[46] To add to the complexity, different components of the JNK pathway can have opposing effects on apoptosis.[47–53] These inconsistencies are due to the fact that the effect of activation of the JNK pathway on apoptosis depends on cell type, the nature of the death stimulus, the duration of its activation and probably, most importantly, the activity of other signaling pathways.

A double-edged sword

Activation of the JNK pathway is involved in apoptosis. The first indication that the JNK pathway might be involved in apoptosis came from studies in neuronal cells. Injection of neutralizing antibodies against c-Jun or overexpression of a c-Jun truncated mutant lacking its N-terminal transactivation domain[54] blocked apoptosis induced by nerve growth factor (NGF) withdrawal. Conversely, overexpression of wild-type c-Jun induced apoptosis of cultured sympathetic neurons.[54] Subsequently, it was reported that apoptosis of rat PC-12 pheochromocytoma cells following NGF withdrawal was inhibited by expressing dominant negative mutants of the JNK pathway, while overexpression of a constitutively active MEKK1, which is a MAP3K for JNK,[29] was able to induce apoptosis in PC-12 cells.[55,56] The results of these experiments suggest a pro-apoptotic role for the JNK pathway in neurons. This conclusion is supported by genetic disruption experiments in mice. Cultured sympathetic neurons of *c-jun*$^{AA63/73}$ mice, in which endogenous *c-jun* alleles were replaced with a non-phosphorylatable *c-jun*$^{AA63/73}$ mutant, turned out to be resistant to apoptosis induced by the excitotoxic glutamate-receptor agonist kainate.[48] In mice deficient in neuronal-specific *jnk3*, hippocampal neurons became resistant to apoptosis induced by kainate[57] and isolated sympathetic neurons were resistant to apoptosis induced by NGF withdrawal.[58] Consistently, in mice deficient in both *jnk1* and *jnk2*, which are ubiquitously expressed, apoptosis was suppressed in the hindbrain neuroepithelium at E9.25 during development.[59,60] The involvement of the JNK pathway in neuronal apoptosis is further supported by the finding that genetic disruption of *jip1*, one of the putative scaffold proteins of the JNK pathway, inhibited JNK activation and c-Jun phosphorylation, resulting in the resistance of hippocampal neurons to kainate-induced apoptosis.[61] Although it is clear that c-Jun-mediated transcription is required for mediating the pro-apoptotic effect of the JNK pathway in neuronal apoptosis, the precise molecular mechanism has yet to be elucidated.

Activation of the JNK pathway has also been implicated in apoptosis of other cell types.[10,11] The most convincing evidence to date comes from mouse embryonic fibroblasts (MEFs) deficient in both *jnk1* and *jnk2*.[49] The resistance of these cells to apoptosis induced by ultraviolet (UV) irradiation suggests that JNK activation is involved in UV-induced apoptosis in MEFs.[49] It will be interesting to find out whether re-expression of JNK1 and JNK2 in these cells can restore the apoptotic response to UV. Unlike in neurons, the pro-apoptotic role of JNK in UV-induced MEF apoptosis appeared to be independent of c-Jun-mediated transcription since it was not affected by the protein synthesis inhibitor cycloheximide or the mRNA synthesis inhibitor actinomycin D.[49] However, MEFs isolated from *c-jun*$^{AA63/73}$ mice are insensitive to UV-induced cell death,[48] suggesting that c-Jun phosphorylation and activation may in fact be required for UV-irradiation-induced cell death. The reason for this discrepancy is currently unknown.

Activation of the JNK pathway is involved in cell survival. Activation of the JNK pathway does not simply serve a pro-apoptotic function. For example, no defect in neuronal apoptosis was evident in *c-jun*$^{-/-}$ mice[47] and many stimuli that activate c-Jun do not induce apoptosis in neurons.[47] In fact, c-Jun is not essential for apoptosis during development since apoptosis takes place in its absence.[62] As with c-Jun, not all stimuli that activate JNK induce apoptosis. Even in the cases where JNK activation contributes to apoptosis, whether it is sufficient to induce cell death is still a matter of debate (see below). Emerging evidence suggests that JNK activation may be required for survival/anti-apoptosis under certain circumstances. In *jnk-1*$^{-/-}$*jnk-2*$^{-/-}$ mice, apoptosis was increased in the hindbrain[59] and forebrain regions[59,60] at E10.5, suggesting that JNK may be required for cell survival in these brain regions during development. In addition, immature thymocytes and peripheral mature T cells deficient in *jnkk1* (*sek1/mkk4*) were highly susceptible to Fas/CD95- and CD3-mediated apoptosis.[63] This suggests that JNKK1 may play an anti-apoptotic role in T cells. Biochemical analysis also revealed that activation of JNK may be involved in suppression of apoptosis in B lymphocytes through phosphorylation and inactivation of the pro-apoptotic molecule Bad.[64] In certain tumor cells, JNK has been shown to function as an anti-apoptotic molecule. Inhibition of JNK activity by specific antisense oligonucleotides (JNKAS) suppressed growth of certain tumor cells through promoting apoptosis.[65,66] Interestingly, the anti-apoptotic function of JNK is related to p53 status since JNKAS inhibited growth of certain p53-deficient, but not p53-positive, tumor cells.[67] It has been reported that activation of the JNK pathway can inhibit p53-induced cell cycle arrest and therefore promote p53-induced apoptosis.[68] This might explain why JNK only exerts its anti-apoptotic function in p53-deficient tumor cells. JNK may also function as an anti-apoptotic molecule in certain tumors in an isoform-specific manner. In *jnk-2*$^{-/-}$ mice, skin papillomas induced by tumor promoter 12-O-tetradecanolyphorbol-13-acetate (TPA) were significantly suppressed due to enhanced apoptosis when compared to wild-type mice.[69] In contrast, TPA-induced skin tumors were enhanced in *jnk-1*$^{-/-}$ mice.[70] The mechanism by which JNK2, but not JNK1, suppresses apoptosis in skin tumors has been attributed to the unique stimulatory effect of JNK2 on AP-1 activity through activation of ERK, rather than c-Jun phosphorylation.[70] Further studies are needed to elucidate the mechanism behind the anti-apoptotic role of JNK.

The observation that the JNK pathway can be either pro-apoptotic or anti-apoptotic suggests that JNK is likely to act as a modulator, rather than as an intrinsic component of the apoptotic machinery. This will be discussed in more detail below.

Conflicting interests

Components of the JNK pathway can have opposing effects on the apoptotic process. Genetic disruption of *jnk1* and *jnk2* in mice caused defects in neuronal apoptosis during development.[49] However, *c-jun*$^{-/-}$ mice do not exhibit this phenotype.[47] This suggests that the pro-apoptotic role of JNK1 and JNK2 in neurogenesis is mediated by a downstream effector other than c-Jun. Another example occurs in mice deficient in *jnkk1* or *c-jun*, which exhibit severe liver tissue degeneration due to apoptosis.[47,53,63] In contrast, this phenotype was not observed in *jnk-1*$^{-/-}$, *jnk-2*$^{-/-}$ or double knockout mice.[49] Interestingly, *c-jun*$^{AA63/73}$ mice display no liver degeneration.[48] These observations suggest that the anti-apoptotic role of JNKK1 in liver may be mediated by downstream effectors other than JNK and phosphorylation of c-Jun is not required for hepatocyte survival. As discussed earlier, JNKK1 protects T cells from apoptosis during their development and activation. In contrast, JNK has been shown to be required for anti-CD3-induced apoptosis in immature thymocytes and is not involved in the apoptosis of mature T cells.[50,51] It appears that, in addition to functioning within the JNK pathway, individual components of the JNK pathway may affect apoptosis through interactions with other cellular signaling effectors.

Family value

It is interesting to note that different JNK isoforms may play distinct roles in apoptosis. *jnk3*$^{-/-}$mice exhibited normal neuronal development,[57] whereas *jnk1*$^{-/-}$*jnk2*$^{-/-}$ mice had defective apoptosis in certain brain regions.[60] However, sympathetic neurons isolated from both mice were resistant to apoptosis induced by environmental stresses such as NGF withdrawal or kainate treatment.[58,60] These observations suggest an involvement of JNK1/JNK2, but not JNK3, in developmental neuronal apoptosis, but with all three JNK isoforms being involved in neuronal apoptosis induced by environmental stresses. It is unclear why the neuronal specific JNK3 is not involved in neuronal apoptosis during development.[57] Since no defects in developmental neuronal apoptosis were detected in *c-jun*$^{-/-}$ mice,[47] the pro-apoptotic effect of JNK1/JNK2 in neuronal development is unlikely to be mediated by c-Jun. Determination of the downstream effector(s) that mediates the pro-apoptotic effect of JNK1/JNK2 during neurogenesis will address this question.

Waiting for the right signals

The nature of the death signal and cell type are critical factors determining whether the JNK pathway is involved in apoptosis. *jnk1*$^{-/-}$*jnk2*$^{-/-}$ MEFs were resistant to apoptosis induced by UV irradiation, anisomycin or MMS, but were still sensitive to anti-Fas antibody.[49] Immature thymocytes deficient in *jnk2* or expressing a non-phosphorylatable JNK1(183A/185F) mutant became resistant to apoptosis induced by anti-CD3 antibody, while remaining sensitive to anti-Fas antibody, UV or dexamethasone-mediated cell death.[50] Furthermore, *jnkk1*$^{-/-}$ immature thymocytes and peripheral mature T cells were highly susceptible to apoptosis mediated by Fas/CD95 and CD3, but not to other stress-related death insults such as UV irradiation, γ-radiation, dexamethasone, anisomycin and heat shock.[63] Thus, although many death signals activate JNK, only a subset of death signals utilizes the JNK pathway to implement apoptosis in a cell-type-dependent manner.

It's all about the network

The fate of a cell following exposure to extracellular stimuli is determined by signal integration of multiple signaling pathways, rather than the activity of a single pathway. It has long been speculated that the role of the JNK pathway in apoptosis is affected by other signaling pathways. This has recently been demonstrated to be the case during TNF-α-induced apoptosis. The proinflammatory cytokine TNF-α activates multiple downstream signaling pathways, including caspases, IKK and JNK.[71] Activation of caspases is required for apoptotic cell death, whereas IKK activation inhibits apoptosis through the transcription factor NF-κB, whose target genes include the inhibitors of apoptosis (IAPs) family.[72] This explains why TNF-α does not typically induce apoptosis.[39] In contrast, biochemical analysis of the role played by JNK activation in TNF-α-induced apoptosis has yielded conflicting results, suggesting pro-apoptotic, anti-apoptotic or no role in this process.[45,46,73] The recent discovery that NF-κB inhibits TNF-α-induced apoptosis by negatively regulating activation of the JNK pathway suggests that the role of the JNK pathway in apoptosis depends on the activity of other signaling pathways.[40,42] JNK activation by TNF-α was shown to be transient in wild-type fibroblasts, but was prolonged in cells deficient in either *Ikkβ*,[40] which is essential for NF-κB activation by TNF-α, or *Rel*A,[40,42] which is a major transactivating subunit of NF-κB. Similar observations were reported using cells expressing a dominant negative IκBα mutant.[41,42] Thus, activation of the NF-κB pathway prevents prolonged JNK activation by TNF-α.[39] Candidate NF-κB-induced JNK inhibitors include X-chromosome-linked IAP (XIAP)[40] and GADD45β.[42] Since genetic disruption of either gene has no significant effect on JNK activation in vivo,[39,74] this suggests that a compensation mechanism may exist and the inhibitory effect of NF-κB may be mediated by multiple proteins. It is important to note that NF-κB-mediated inhibition may target a unique component(s) in the TNF-α-activated JNK pathway since it does not inhibit JNK activation by interleukin (IL)-1[40] or UV (Y. Minemoto, B.C. Dibling and A. Lin, unpublished results). Importantly, abrogation of prolonged JNK activation by expressing a dominant negative JNKK2(K149M) mutant[40] or using a specific JNK inhibitor SP600125 (G. Tang, Y. Minemoto, F. Tang and A. L., unpublished results) resulted in suppression of TNF-α-induced apoptosis in $RelA^{-/-}$ or $Ikk\beta^{-/-}$ fibroblasts. Taken together, the role of NF-κB in prevention of TNF-α-induced apoptosis is two-fold: to block both caspase activation and prolonged JNK activation. In support of the idea that prolonged JNK activation contributes to TNF-α-induced apoptosis, it was found that JNK activation by TNF-α was transient in TNF-α-resistant human breast carcinoma MCF-7-R cells, but was prolonged in TNF-α-sensitive MCF-7 cells.[75] Restoration of transient JNK activation in MCF-7 cells inhibited TNF-α-induced apoptosis.[75] These observations also explain why transient activation of the JNK pathway by TNF-α does not contribute to TNF-α-induced apoptosis.[45,46]

Time matters

The duration of JNK activation affects its role in apoptosis. A correlation between prolonged JNK activation and cell death induced by UV or γ-radiation has been observed.[76] Furthermore, inhibition of tyrosine phosphatase(s) by inhibitors such as vanadate resulted in prolonged JNK activation and sensitized cells to TNF-α-induced apoptosis.[43] However, it is not clear whether prolonged JNK activation causes or is a consequence of apoptosis, since the JNK pathway can also be activated by caspases.[77] As discussed earlier, prolonged but not transient JNK activation contributes to TNF-α-induced apoptosis in $RelA^{-/-}$ or $Ikk\beta^{-/-}$ fibroblasts[40] (G. Tang, Y. Minemoto, F. Tang and A. L., unpublished results) and TNF-α-sensitive MCF-7 cells.[75] However, prolonged JNK activation alone may not be sufficient to induce apoptosis. Expression of the constitutively active JNKK2-JNK1 fusion protein, which exerts prolonged JNK activity, does not induce apoptosis in $RelA^{-/-}$ fibroblasts[40] or other cells (G. Tang, Y. Minemoto, F. Tang and A. Lin, unpublished results). It is likely that prolonged JNK activation may only promote apoptosis in cells where the apoptotic program has been initiated.

It has yet to be determined at what point the JNK pathway is targeted by NF-κB-induced inhibitors. JNK itself is unlikely to be the target of NF-κB-induced inhibition since the activity of a constitutively active JNK construct, the JNKK2–JNK1 fusion protein,[40,78] is not affected by NF-κB-induced inhibition (G. Tang, Y. Minemoto, F. Tang and A. L., unpublished results). It is likely that NF-κB-induced inhibitor(s) may target the component(s) upstream of JNK in the TNF-α signaling pathway.

The mechanism by which prolonged but not transient JNK activation promotes TNF-α-induced apoptosis is currently unknown. One possibility is that prolonged JNK activation may lead to sustained c-Jun-mediated transcription of genes that are involved in apoptosis. This is unlikely since TNF-α was still able to induce apoptosis in $Ikk\beta^{-/-}$ fibroblasts in the presence of the protein synthesis inhibitor cycloheximide (CHX) (G. Tang, Y. Minemoto, F. Tang and A. L., unpublished results). JNK activation has been reported to be required for activation of the mitochondrial-dependent death pathway in UV-induced apoptosis.[49] It is interesting to note that the inhibition of prolonged JNK activation blocks UV-irradiation-induced apoptosis,[49] which is mainly mediated by the mitochondrial-dependent death pathway, but only suppresses TNF-α-induced apoptosis[40] (G. Tang, Y. Minemoto, F. Tang and A. L., unpublished results), which is mediated by both receptor- and mitochondrial-dependent death pathways. This suggests that in the absence of NF-κB activation, prolonged JNK activation may have a pro-apoptotic role in TNF-α-induced apoptosis through the mitochondrial-death pathway.

How does JNK activation contribute to apoptosis?

Despite overwhelming evidence pointing to the involvement of the JNK pathway in apoptosis following certain stimuli, the molecular mechanism behind JNK's pro- or anti-apoptotic action remains unclear. Two hypotheses have recently been proposed to explore how activation of the JNK pathway contributes to the apoptotic process. The key issue is whether JNK is an intrinsic component or a modulator of the death pathway, as discussed below.

The ''activating the mitochondrial-dependent apoptotic pathway'' hypothesis

It was hypothesized that JNK is an intrinsic component of the mitochondrial-dependent death pathway that mediates stress-induced apoptosis.[49] During the apoptotic process, cytochrome C is released from mitochondria and co-operates with Apaf-1 to activate initiator caspase 9.[7] The release of cytochrome C is also associated with changes in mitochondrial membrane potential, a critical index of activation of the mitochondrial-dependent death pathway.[7] *jnk1*$^{-/-}$*jnk2*$^{-/-}$ MEFs were resistant to apoptosis induced by stresses such as UV irradiation, anisomycin and MMS. This is most likely the result of failure to activate the mitochondrial-dependent death pathway in these cells, since UV was unable to induce cytochrome C release or depolarization of mitochondrial membrane potential.[49] In contrast, the receptor-dependent death pathway was not affected, as anti-Fas antibody still induced apoptosis in these cells.[49] It is clear that activation of the JNK pathway contributes to UV-induced apoptosis through the mitochondrial-dependent death pathway.[49] However, whether activation of JNK is sufficient to induce apoptosis is debatable. If JNK is an intrinsic component of the death pathway, activation of JNK alone should be able to induce death. In support of this idea, it has recently been reported that, in CHO cells, the constitutively active JNKK2–JNK1 fusion protein was sufficient to induce apoptosis through activation of the mitochondrial-dependent death pathway.[79] This contradicts the finding that expression of the constitutively active JNKK2–JNK1 fusion protein does not induce apoptosis in *RelA*$^{-/-}$ fibroblasts.[40] The constitutively active JNK also does not induce death in other cells including wild-type or *Ikkβ*$^{-/-}$ fibroblasts and HeLa cells (G. Tang, Y. Minemoto, F. Tang and A. L., unpublished results). Furthermore, there is no increased caspase activation or DNA fragmentation in cells expressing the constitutively active JNK when compared to cells expressing a control vector or a kinase-deficient JNKK2(K149M)-JNK1 mutant construct (G. Tang, Y. Minemoto, F. Tang and A. Lin, unpublished results). Identification of the downstream target(s) that mediates the pro-apoptotic effect of JNK should resolve this discrepancy. The downstream target protein(s) of JNK during apoptosis is presently unknown. It has been shown that JNK can phosphorylate and inactivate Bcl-2 and Bcl-XL, which are both negative regulators of mitochondrial cytochrome C release.[21,22] However, there is no evidence that Bcl-2 or Bcl-XL were phosphorylated by JNK in UV-irradiated MEFs.[49] Further studies are needed to determine whether JNK activation is sufficient to induce or only promote apoptosis.

The ''breaking the brake on apoptosis'' hypothesis

We envision that JNK may be a modulator rather an intrinsic component of the apoptotic machinery. Thus, JNK activation facilitates but does not induce apoptosis. Specifically, we hypothesize that activated JNK inactivates suppressors of the apoptotic machinery, thereby ''breaking the brake'' on apoptosis. According to this model, JNK activation contributes to apoptosis only if the apoptotic process has already been activated. This can explain how prolonged JNK activation alone does not induce apoptosis, but is able to promote TNF-α-induced apoptosis in the absence of NF-κB activation (Fig. 2). As discussed earlier, TNF-α is not typically a killer since activation of NF-κB inhibits caspases and prevents prolonged JNK activation.[39,40] In the absence of NF-κB-mediated inhibition, such as in *RelA*$^{-/-}$ fibroblasts, TNF-α induces apoptosis through activation of caspases and release of apoptotic pathway inhibition by prolonged JNK activation (Fig. 2). This model may also explain how activation of JNK alone does not induce apoptosis, but inhibition of JNK activation blocks UV-induced cell death. UV induces apoptosis by activating the mitochondrial-dependent death pathway and it also induces prolonged JNK activation by an unknown mechanism.[49] Nevertheless, prolonged JNK activation by UV may function to inactivate inhibitors of the mitochondrial-dependent death pathway, thereby contributing to activation of the death pathway. It is interesting to note that inhibition of prolonged JNK activation only suppresses TNF-α-induced apoptosis,[40] but completely blocks UV-induced apoptosis.[49] Considering TNF-α activates both receptor- and mitochondrial-dependent death pathways, while UV activates only the mitochondrial-dependent death pathway, so it is likely that prolonged JNK activation inactivates inhibitors involved in blocking the mitochondrial-dependent death pathway. Thus, the role of the JNK pathway in apoptosis can be thought of as being analogous to driving a car. If you just start the engine (activation of apoptotic pathways) or release the brake (prolonged activation of JNK), the car (apoptosis) will not move. The car (apoptosis) will move only when you start

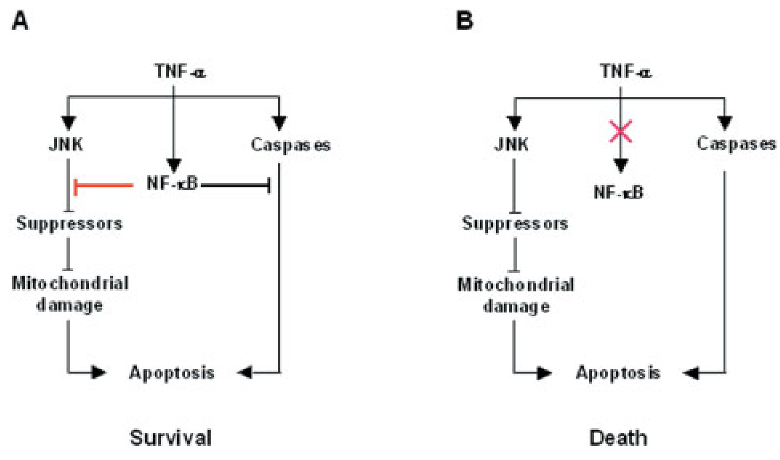

Figure 2. The "breaking the brake" on apoptosis model that explains the role of JNK activation in TNF-α-induced apoptosis. **A:** While caspase activation initiates and executes apoptosis, prolonged JNK activation promotes apoptosis by inactivating suppressors of the mitochondrial-dependent death pathway, i.e., breaking the brake on apoptosis. Activation of NF-κB by TNF-α blocks caspase activation and prevents prolonged JNK activation, thereby inhibiting TNF-α-induced apoptosis. **B:** Inactivation of NF-κB removes inhibition on both caspases and JNK, allowing TNF-α to induce apoptosis.

the engine (activation of apoptotic pathways) and release the brake (prolonged activation of JNK). It remains to be determined whether this "breaking the brake" on apoptosis model can be used to explain the role of the JNK pathway in other types of apoptosis.

Concluding remarks

The role of the JNK pathway in apoptosis is complex, as it is able to promote, suppress or have no role in the process. Although activation of the JNK pathway can contribute to apoptosis, JNK may not be an intrinsic component of the apoptotic machinery. Instead, activation of JNK may modulate the apoptotic process in a cell-type and stimulus-dependent manner. The contribution of the JNK pathway to TNF-α-induced apoptosis is determined by NF-κB-mediated negative regulation. It remains to be determined whether this is true in apoptosis induced by other death signals. Future studies will elucidate how NF-κB-induced inhibitors block activation of the JNK pathway and how prolonged but not transient JNK activation contributes to apoptosis. Since deregulation of the JNK pathway has been implicated in cancer[69,80,81] and other diseases, including Alzheimer's and Parkinson's disease,[82,83] heart hypertrophy and ischemia,[84–86] investigation of the molecular mechanisms that govern the role of the JNK pathway in apoptosis should provide insight into its biological functions and strategies to target this pathway for prevention and treatment of human disease and cancer.

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